

Course Title: An Introduction to Bayesian Analysis
Course code: MTH422A
Semester: Spring
Academic Year: 2023-2024
Start Date: 04/01/2023
Class Schedule: Monday, Wednesday, Friday 9:00 AM - 10:00 AM
Lab Schedule: Thursday 5:00 PM - 6:00 PM
Location: L10, NCL

Instructor: Arnab Hazra
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Course Purpose

In this course, the students will be introduced to the Bayesian approach to statistics. We will start with some preliminary concepts about probability theory, and then we will gradually move toward data analysis. Students will learn the main philosophical difference between the commonly-taught Frequentist approach and the Bayesian paradigm. They will see some benefits of using a Bayesian approach, in terms of better uncertainty quantification and obtaining results that are more intuitive and more interpretable.

Learning Outcomes

After completing this course, students will learn the concepts of the Bayesian approach, the key differences between the Bayesian and Frequentist paradigms, and different types of priors, and they will have the ability to do basic data analyses.

Prerequisites

1. Basic probability theory
2. Some algebra and calculus
3. Basic visualization techniques
4. Statistical computing using R/JAGS

Materials

Lecture notes/slides will be provided. Additionally, students can read the following books:

1. *Bayesian Statistical Methods* by Brian J. Reich, Sujit K. Ghosh. Published in 2019, by Chapman and Hall/CRC (I will be mainly following this book).
2. *Bayesian Essentials with R* by Jean-Michel Marin and Christian P. Robert. Published in 2014, by Springer.
3. *Bayesian Data Analysis* by Andrew Gelman, John Carlin, Hal Stern, David Dunson, Aki Vehtari, and Donald Rubin. Published in 2013 by Chapman and Hall/CRC. For non-commercial purposes, a free electronic copy can be downloaded from <http://www.stat.columbia.edu/~gelman/book/>.
4. *A First Course in Bayesian Statistical Methods* by Peter D. Hoff. Published in 2009, by Springer.
5. *An Introduction to Bayesian Analysis* by Jayanta K. Ghosh, Mohan Delampady, Tapas Samanta. Published in 2006 by Springer (printed Indian edition available).

Grading

Assignments: 5 X 10 = 50 (both analytical as well as coding)

Mid-semester exam: 25 (only analytical)

End-semester exam: 25 (only analytical)

Tentative schedule

- Basics of Bayesian paradigm, with some examples
- A brief review of probability theory
- Bayes' theorem and its implications, some examples
- Introduction to Bayesian inference
- Summarizing a posterior
- Conjugate priors part 1
- Conjugate priors part 2
- Objective Bayes
- Bayesian computing: deterministic methods
- Markov chain Monte Carlo part 1
- Markov chain Monte Carlo part 2
- Diagnosing and improving convergence
- Questions and answers discussions
- Bayesian linear models part 1
- Bayesian linear models part 2
- Advanced Bayesian modeling part 1
- Advanced Bayesian modeling part 2
- Paper presentation
- Model comparisons part 1
- Model comparisons part 2
- Hierarchical models part 1
- Hierarchical models part 2
- Case study demonstration part 1
- Case study demonstration part 2
- Frequentist properties of Bayesian methods, decision theory

Active learning

To create an active learning experience, this course combines lectures, computer demonstrations, readings, discussions, and paper and project presentations.

Policy on Academic Integrity

Any cheating during the examinations and assignments will be considered a serious offense and the involved students will be penalized substantially.

Course structure

The course structure is inspired by “A Bayesian Statistics Course for Undergraduates: Bayesian Thinking, Computing, and Research” by Jingchen Hu (2020), *Journal of Statistics Education*, 28:3, 229-235.

Some of the course contents are influenced by the courses *Applied Bayesian Analysis* (MS level) taught by Brian J. Reich, at North Carolina State University, United States.

Grading policy (for all students): Relative. A slightly easier grading will be opted for BS SDS students as it is a compulsory course for them.